

# Immunodeficiency 105 (IMD105 / CD45 Deficiency SCID): Comprehensive Disease Characteristics Report

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**Disease:** Immunodeficiency 105 · **MONDO:** MONDO:0800104 · **OMIM:** #619924 (phenotype), 151460 (gene) · **Gene:** *PTPRC* (CD45) · **Category:** Mendelian (autosomal recessive)

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## Summary

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**Immunodeficiency 105 (IMD105) is an ultra-rare autosomal recessive severe combined immunodeficiency (SCID) caused by biallelic loss-of-function mutations in *PTPRC*, the gene encoding the leukocyte common antigen CD45.** CD45 is an abundant, hematopoietic-specific, receptor-type transmembrane protein tyrosine phosphatase (UniProt P08575; EC 3.1.3.48) expressed on all nucleated blood cells. Its principal function is to activate the Src-family kinases Lck and Fyn — by dephosphorylating their inhibitory C-terminal tyrosine — which are indispensable for T-cell-receptor (TCR) and B-cell-receptor (BCR) proximal signaling. When CD45 is absent, Lck remains locked in an inhibited conformation, antigen-receptor signaling collapses, and thymic T-cell development arrests. The clinical result is a **T-cell-negative, B-cell-positive, NK-variable (T–B+NK±) SCID** that presents in early infancy (~2 months) with severe, recurrent, opportunistic infections and progressive hypogammaglobulinemia despite normal B-cell numbers.

The disease was defined by only a small number of human patients. The first (Kung et al., 2000) carried a large deletion on one *PTPRC* allele plus an IVS13 splice-site mutation on the other; the second (Tchilian et al., 2001) was homozygous for a 6-bp in-frame deletion removing Glu339/Tyr340 in the first fibronectin type III module of the CD45 extracellular domain, abolishing surface CD45. A *Ptprc*-null mouse faithfully recapitulates the human immunophenotype (T-low, B-normal-to-high) with a two-stage block in thymic T-cell development, providing strong mechanistic corroboration. Importantly, the common *PTPRC* **C77G isoform-altering polymorphism** — associated with autoimmune and infectious-disease susceptibility — is mechanistically and clinically **separate** from the biallelic null mutations that cause IMD105.

Diagnosis begins with universal newborn screening for T-cell receptor excision circles (TRECs); affected infants are T-lymphopenic with low/absent TRECs. Confirmation relies on lymphocyte immunophenotyping (T–B+NK± with the diagnostically distinctive **absence of surface CD45 on all leukocytes**), absent mitogen proliferation, and molecular sequencing of *PTPRC*. The only curative therapy is **allogeneic hematopoietic stem cell transplantation (HSCT)**; untreated disease is fatal in early childhood. Survival is markedly better when disease is detected pre-symptomatically by newborn screening. This report compiles the etiology, phenotype spectrum, molecular mechanism, anatomy, epidemiology, diagnostics, prognosis, treatment, prevention, and model-organism data with supporting ontology terms and primary-literature citations.

## Key Findings

### Finding 1 — IMD105 is CD45 deficiency SCID caused by biallelic *PTPRC* mutations

IMD105 (OMIM #619924; MONDO:0800104) is a severe combined immunodeficiency caused by homozygous or compound-heterozygous loss-of-function mutations in *PTPRC* (CD45; **HGNC:9666**; NCBI Gene 5788) on chromosome **1q31.3**. Inheritance is autosomal recessive. The immunophenotype is **T– B+ NK-variable SCID** with onset of recurrent infections in early infancy. The disorder sits within the OMIM Autosomal Recessive SCID Phenotypic Series (PS601457).

The first CD45-deficient patient (Kung et al., 2000) presented at 2 months with SCID, carrying a large deletion on one allele and an IVS13 donor splice-site point mutation on the other; peripheral T cells were greatly diminished and mitogen-unresponsive, B-cell numbers were normal, but serum immunoglobulins declined with age. The second patient (Tchilian et al., 2001) was homozygous for a 6-bp deletion causing loss of Glu339 and Tyr340 in the first fibronectin type III module of the extracellular domain, which prevented CD45 surface expression.

"The patient presented at 2 months of age with severe combined immunodeficiency disease." — [PMID: 10700239](#)

"Thus, CD45 deficiency in humans results in T- and B-lymphocyte dysfunction." — [PMID: 10700239](#)

"a homozygous 6-bp deletion in the gene encoding CD45 (PTPRC, gene map locus 1q31-32), which results in a loss of glutamic acid 339 and tyrosine 340 in the first fibronectin type III module of the extracellular domain of CD45, is associated with failure of surface expression of CD45 and SCID" — [PMID: 11145714](#)

## Finding 2 — Mechanism: CD45 activates Src-family kinases (Lck/Fyn) required for antigen-receptor signaling

CD45 is an abundant, hematopoietic-specific transmembrane tyrosine phosphatase on all leukocytes. It dephosphorylates the inhibitory C-terminal tyrosine of Lck (Y505) to relieve autoinhibition and permit the active, open conformation. The phospho-balance at Lck's inhibitory versus activating tyrosines is set jointly by CD45 (activating) and Csk (inhibitory). Loss of CD45 therefore leaves Lck hyperphosphorylated and inactive, abrogating TCR- and BCR-proximal signaling and blocking thymocyte and T-cell development. CD45 additionally has a *negative* role — it can dephosphorylate the TCR itself — and its spatial exclusion from TCR clusters tunes the net signaling outcome (reconstituted-membrane studies).

"The hematopoietic-specific transmembrane protein tyrosine phosphatase CD45 functions to regulate Src kinases required for T- and B-cell antigen receptor signal transduction." — [PMID: 10700239](#)

"The balance of phosphorylation at the inhibitory and activating Tyr residues is maintained by a balance between CD45 and Csk" — [PMID: 32794043](#)

"CD45 is an abundant transmembrane tyrosine phosphatase, expressed on all leukocytes, and is required for efficient lymphocyte signaling." — [PMID: 11145714](#)

## Finding 3 — CD45-null mouse recapitulates human T–B+ immunophenotype (model organism)

*Ptprc* (Cd45) knockout mice (Byth et al., 1996) completely lack all CD45 isoforms and display an immunophenotype closely matching the human disease: ~5-fold reduction in splenic T cells and ~2-fold increase in B cells versus controls. T-cell development is blocked at two stages — a reduced double-negative→double-positive (DN→DP) transition (~2-fold) and a severely

impaired DP→single-positive maturation (~4–5-fold). CD45-null thymocytes are severely impaired in TCR-crosslinking-induced apoptosis (negative selection) yet respond normally to non-TCR signals, and show defective superantigen (SEB)-mediated deletion. CD45 is expressed highest on large, cycling, positively-selected DP thymocytes and enhances positive selection. Mouse *Ptpnc* = NCBI Gene 19264, ortholog of human *PTPRC* (NCBI Gene 5788).

"The spleens from CD45-null mice contain approximately twice the number of B cells and one fifth the number of T cells found in normal controls." — [PMID: 8666928](#)

"T cell development is significantly inhibited in CD45-null animals at two distinct stages." — [PMID: 8666928](#)

#### **Finding 4 — *PTPRC* C77G is a distinct isoform-altering susceptibility polymorphism, NOT a cause of IMD105**

The **C77G polymorphism** in *PTPRC* exon 4/exon A disrupts an exonic splicing silencer (ESS1/ARS motif), preventing hnRNP L/K/E2-mediated repression and causing aberrant retention of CD45RA isoforms on activated/memory T cells (Motta-Mena 2011). C77G occurs at low frequency in healthy individuals but is enriched in multiple sclerosis, systemic sclerosis, autoimmune hepatitis, hepatitis C, and HIV-1 cohorts; it enhances TCR signaling and alters cytokine (IL-2/Jak1) responsiveness and adhesion. A second isoform-altering polymorphism, A138G, is associated with Graves' disease. The MS association is inconsistent across populations (no association in an Australian cohort). This is a partial gain/alteration-of-function *susceptibility allele* — mechanistically and clinically **separate** from the biallelic loss-of-function that causes IMD105 SCID. This distinction is essential for correctly curating the disease entry.

"An enhanced frequency of C77G individuals has been reported in cohorts of patients suffering from multiple sclerosis, systemic sclerosis, autoimmune hepatitis, hepatitis C and human immunodeficiency virus (HIV)-1." — [PMID: 17903220](#)

"the C77G polymorphism, which correlates with autoimmune disease susceptibility in humans, disrupts exon silencing by preventing the redundant activity of hnRNPs K and E2 to compensate for the weakened function of hnRNP L" — [PMID: 21507955](#)

"Two polymorphisms (C77G and A138G), which alter CD45 isoform expression, are associated with autoimmune and infectious diseases." — [PMID: 18312479](#)

## Finding 5 — Diagnostics: TREC newborn screening detects CD45-deficiency SCID; confirmed by flow cytometry and *PTPRC* sequencing

SCID, including CD45 deficiency, is detectable at birth by quantifying T-cell receptor excision circles (TRECs) in dried blood spots; CD45-deficient infants are T-lymphopenic and will have low/absent TRECs. In the Catalonia program (2017–2023), among 420,263 newborns, 105 screened positive (0.02%) using the EnLite Neonatal TREC assay (cut-off 20 copies/ $\mu$ L), yielding an overall SCID incidence of **1:46,753 live births**. Combined TREC/KREC assays additionally flag B-cell lymphopenia. Confirmatory workup: lymphocyte immunophenotyping by flow cytometry (T<sup>+</sup>/B<sup>+</sup>/NK-variable; **absent surface CD45 on all leukocytes is diagnostic and distinctive**), mitogen proliferation assays (absent), serum immunoglobulins (hypogammaglobulinemia), and molecular confirmation by *PTPRC* sequencing (single-gene, gene panels, or WES/WGS).

"Severe combined immunodeficiency (SCID) can be detected at birth through T-cell receptor excision circles (TREC) analysis in dried blood spots." — [PMID: 42079620](#)

"is associated with failure of surface expression of CD45 and SCID" — [PMID: 11145714](#)

## Finding 6 — Treatment & prognosis: Allogeneic HSCT is the only curative therapy; early diagnosis markedly improves survival

Allogeneic hematopoietic stem cell transplantation (HSCT/HCT) is the **only curative treatment** for SCID including CD45 deficiency; without immune reconstitution, death occurs in early childhood. Outcome is strongly time-dependent. In a Brazilian cohort, 2-year overall survival was **71.4% for early-diagnosed** (newborn-screened/family history) versus **29.2% for late clinically-diagnosed** SCID ( $p = 0.053$ ). In a 100-patient single-center HSCT cohort, overall survival was 68% (84% excluding first-month mortality), and NBS-identified cases had superior OS (93%) versus clinically-identified cases ( $p = 0.04$ ); better outcomes were associated with bone-marrow stem-cell source, matched related donors, and use of conditioning. Supportive care before transplant: protective isolation, antimicrobial/antifungal/anti-*Pneumocystis* prophylaxis, immunoglobulin replacement, avoidance of live vaccines, and use of irradiated/CMV-safe/leukoreduced blood products. Gene therapy is investigational for other SCID subtypes but not established for *PTPRC*.

"Hematopoietic cell transplantation (HCT) is the only curative treatment currently available in Brazil." — [PMID: 40374985](#)

"The 2-year overall survival (OS) of the late group was 29.2%, in contrast to the 2-year OS of the early diagnosis group of 71.4%" — [PMID: 40374985](#)

"SCID cases identified by NBS demonstrated superior OS (93%) compared to cases identified by clinical presentation" — [PMID: 39900265](#)

## Finding 7 — Clinical phenotype spectrum with HPO terms

Per the OMIM #619924 clinical synopsis and the reported patients: onset in early infancy (~2 months) with recurrent/severe infections. Manifestations include recurrent respiratory infections/pneumonia, severe combined immunodeficiency, T-cell lymphopenia (decreased/absent nonfunctional T cells), normal-to-increased B cells (B+), hypogammaglobulinemia that worsens with age, absent lymphocyte proliferative response to mitogens, dermatitis/eczema, lymphadenopathy, and B-cell lymphoma in one patient. NK cells are normal or low. Severity is severe; the course is rapidly life-threatening without HSCT; expressivity is somewhat variable across the few cases. Typical opportunistic organisms include *Pneumocystis jirovecii*, CMV, and other viral/fungal pathogens.

| Phenotype                                            | HPO term                   | Frequency/notes      |
|------------------------------------------------------|----------------------------|----------------------|
| Severe combined immunodeficiency                     | HP:0004430                 | Defining feature     |
| Reduced T-cell count / T lymphopenia                 | HP:0410354 /<br>HP:0005415 | Core; T–             |
| Pneumonia                                            | HP:0002090                 | Recurrent            |
| Recurrent respiratory infections                     | HP:0002205                 | Frequent             |
| Decreased circulating IgG level                      | HP:0004315                 | Progressive with age |
| Decreased circulating IgA level                      | HP:0002720                 | Variable             |
| Abnormal T cell proliferation (mitogen-unresponsive) | HP:0031381                 | Consistent           |
| Eczema / inflammatory skin abnormality               | HP:0000964 /<br>HP:0011123 | Reported             |
| Lymphadenopathy                                      | HP:0002716                 | Reported             |
| B-cell lymphoma                                      | HP:0012191                 | 1 patient            |

"The population of peripheral blood T lymphocytes was greatly diminished and unresponsive to mitogen stimulation." — [PMID: 10700239](#)

"Despite normal B-lymphocyte numbers, serum immunoglobulin levels decreased with age." — [PMID: 10700239](#)

## Finding 8 — Gene/protein annotation: *PTPRC*/CD45 (UniProt P08575) with GO terms

Causal gene *PTPRC* (**HGNC:9666**; NCBI Gene 5788; Ensembl ENSG00000081237; chromosome 1q31.3; OMIM gene 151460). Protein: Receptor-type tyrosine-protein phosphatase C (CD45, leukocyte common antigen), **UniProt P08575**, EC 3.1.3.48. It is a single-pass type I transmembrane protein with a large N-glycosylated extracellular domain (residues 26–577; two fibronectin type-III domains at 391–483 and 484–576), a transmembrane helix (578–598), and a large cytoplasmic region (599–1306) containing two tandem PTP domains — a membrane-proximal, catalytically active D1 (653–912) and a membrane-distal, regulatory/



inactive D2 (944–1228). Alternative splicing of exons 4/5/6 (A/B/C) generates the CD45RA, RB, RC, and RO isoforms. Disease-causing variants are germline loss-of-function (large deletion + splice IVS13; homozygous in-frame 6-bp deletion Glu339\_Tyr340del) with autosomal recessive inheritance and complete penetrance.

**Suggested GO annotations:** - **Molecular Function:** transmembrane receptor protein tyrosine phosphatase activity (GO:0005001); protein tyrosine phosphatase activity (GO:0004725); protein tyrosine kinase inhibitor activity (GO:0030292) - **Biological Process:** T cell receptor signaling pathway (GO:0050852); positive regulation of antigen receptor-mediated signaling pathway (GO:0050857); T cell differentiation (GO:0030217); T cell activation (GO:0042110); B cell receptor signaling pathway (GO:0050853); B cell differentiation (GO:0030183); natural killer cell differentiation (GO:0001779); DN2 thymocyte differentiation (GO:1904155); protein dephosphorylation (GO:0006470) - **Cellular Component:** external side of plasma membrane (GO:0009897); membrane raft (GO:0045121); plasma membrane (GO:0005886)

"a loss of glutamic acid 339 and tyrosine 340 in the first fibronectin type III module of the extracellular domain of CD45, is associated with failure of surface expression of CD45" — PMID: [11145714](#)

## Finding 9 — Anatomical structures, cell types, and subcellular localization (UBERON / CL / GO-CC)

CD45 is expressed on all nucleated hematopoietic cells (leukocyte common antigen). The primary affected system is the immune/hematolymphoid system (**UBERON:0002405** immune system; **UBERON:0002390** hematopoietic system). Primary organs/tissues: **thymus (UBERON:0002370)** — the site where T-cell development arrests at the DP→SP transition; **bone marrow (UBERON:0002371)** — hematopoietic origin; **spleen (UBERON:0002106)**; **lymph nodes (UBERON:0000029, clinical lymphadenopathy)**; **peripheral blood (UBERON:0000178)**. Secondary/complication organs: **lung (UBERON:0002048; pneumonia)** and **skin (UBERON:0002097; dermatitis)**.

**Cell types (Cell Ontology):** T cell (CL:0000084), thymocyte (CL:0000893), CD4/CD8 double-positive thymocyte (CL:0000807), mature  $\alpha\beta$  T cell (CL:0000791), B cell (CL:0000236), natural killer cell (CL:0000623), leukocyte/hematopoietic cell (CL:0000738).

**Subcellular localization (GO-CC):** plasma membrane (GO:0005886), external side of plasma membrane (GO:0009897), membrane raft/microdomain (GO:0045121, GO:0098857) where CD45 regulates raft-associated Src kinases.



"T cell development is significantly inhibited in CD45-null animals at two distinct stages."  
— [PMID: 8666928](#)

"CD45 is an abundant transmembrane tyrosine phosphatase, expressed on all leukocytes"  
— [PMID: 11145714](#)

## Finding 10 — Inheritance, epidemiology, and prevention

Inheritance is **autosomal recessive with complete penetrance** for biallelic loss-of-function; heterozygous carriers are clinically unaffected. IMD105 is **ultra-rare** — only ~3 unrelated patients are reported in OMIM — so no disease-specific prevalence/incidence exists; it is a molecularly rare subtype within the broader SCID category (overall SCID incidence ~1:46,753 to ~1:58,000 live births by TREC screening). Consanguinity/homozygosity is a risk context (e.g., the homozygous 6-bp deletion). No sex predilection is expected (autosomal). There is no environmental cause; infections are downstream consequences (opportunistic *Pneumocystis jirovecii*, CMV, other viruses/fungi), not triggers.

**Prevention/management:** (1) secondary prevention via universal newborn TREC screening for presymptomatic detection; (2) genetic counseling for recurrence risk (25% for carrier couples), carrier testing of relatives, and prenatal/preimplantation genetic testing when the familial *PTPRC* variants are known; (3) tertiary prevention of complications pre-HSCT — protective isolation, antimicrobial/anti-*Pneumocystis* prophylaxis, IVIG replacement, only irradiated/CMV-safe/leukoreduced blood products, and strict avoidance of live vaccines (BCG, rotavirus, MMR, VZV, OPV).

"corresponding to an overall incidence of 1:46,753 live births" — [PMID: 42079620](#)

"a homozygous 6-bp deletion in the gene encoding CD45" — [PMID: 11145714](#)

## Section-by-Section Report

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### 1. Disease Information

IMD105 is an autosomal recessive severe combined immunodeficiency of the T–B+NK-variable type caused by complete deficiency of the leukocyte common antigen CD45 (gene *PTPRC*). It presents in early infancy with life-threatening opportunistic infections. **Key identifiers:** OMIM #619924 (phenotype), OMIM 151460 (gene); MONDO:0800104; part of OMIM Phenotypic Series PS601457 (autosomal recessive SCID). ICD-10 D81.x (combined immunodeficiencies); ICD-11 4A01.0Y (other combined immunodeficiencies). MeSH: Severe Combined Immunodeficiency (D016511); no CD45-specific MeSH descriptor. **Synonyms:** CD45 deficiency; SCID due to CD45 deficiency; leukocyte common antigen deficiency; *PTPRC* deficiency; T-cell-negative, B-cell-positive SCID due to CD45 deficiency. The information is derived from **aggregated disease-level resources** (OMIM, primary case reports) rather than EHR — the entire literature rests on ~3 individually reported patients plus mouse-model data.

### 2. Etiology

**Causal factor:** purely genetic — biallelic loss-of-function mutations in *PTPRC* (chromosome 1q31.3). **Genetic risk factors:** the only causal factor is inheriting two damaging *PTPRC* alleles; consanguinity increases homozygosity risk. There are **no environmental risk factors** for disease causation. The recurrent opportunistic infections (*Pneumocystis jirovecii*, CMV, other viruses/fungi) are downstream consequences of immune failure, not triggers. **Protective factors:** none genetic beyond simply not carrying biallelic null alleles. **Gene–environment interactions:** not applicable to disease causation; environment determines which opportunistic infections manifest. Note: the *PTPRC* **C77G** and A138G polymorphisms are isoform-altering *susceptibility* alleles for autoimmune/infectious diseases and are **not** part of IMD105 etiology (Finding 4).

### 3. Phenotypes

See Finding 7 table. Phenotype **onset is neonatal-to-early-infancy** (~2 months); **severity is severe; progression is progressive and rapidly life-threatening** without HSCT. Laboratory abnormalities (T lymphopenia, absent mitogen response, progressive hypogammaglobulinemia, absent surface CD45) are the most consistent features. Quality-of-life

impact is profound: without treatment the disease is uniformly fatal in early childhood; with successful HSCT, survivors may achieve durable immune reconstitution. Behavioral phenotypes are not a feature.

#### 4. Genetic / Molecular Information

**Causal gene:** *PTPRC* (HGNC:9666; NCBI Gene 5788; Ensembl ENSG00000081237; OMIM 151460).

**Pathogenic variants reported:** (i) large deletion on one allele + IVS13 donor splice-site point mutation (compound heterozygous; Kung 2000); (ii) homozygous in-frame 6-bp deletion p.Glu339\_Tyr340del in the first fibronectin type III module (Tchilian 2001). **Variant classes:** structural (large deletion), splice-site, and in-frame indel — all loss-of-function. **Functional consequence:** loss of function → absent or non-functional surface CD45. **Classification:** pathogenic per ACMG (null/LOF in a gene with established LOF mechanism, segregating with recessive disease). **Allele frequency:** the disease alleles are private/ultra-rare (absent from gnomAD at appreciable frequency). **Origin:** germline. **Modifier genes:** none established. **Epigenetics/chromosomal abnormalities:** none specific to IMD105 (though the causal lesions include a submicroscopic deletion detectable in principle by high-resolution methods).

#### 5. Environmental Information

No environmental, lifestyle, toxicological, or radiation factors cause IMD105. **Infectious agents** are relevant only as downstream opportunistic complications — *Pneumocystis jirovecii*, cytomegalovirus, and other viral/fungal/opportunistic pathogens typical of SCID. Live vaccine organisms (e.g., BCG, rotavirus, OPV) pose iatrogenic infection risk and must be avoided.

#### 6. Mechanism / Pathophysiology

**Ordered causal chain (initiating lesion → clinical manifestation):**

1. Biallelic loss-of-function *PTPRC* mutation **leads to** absent or non-functional CD45 protein at the leukocyte surface (demonstrated: absent surface CD45;

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2. Absence of CD45 phosphatase activity **results in** failure to dephosphorylate the inhibitory C-terminal tyrosine (Y505) of the Src-family kinase Lck (and Fyn) (demonstrated biochemically;

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3. Persistent inhibitory phosphorylation **causes** Lck to remain in its closed, autoinhibited conformation, so active Lck is not generated (mechanistic, in vitro/reconstitution evidence).
4. Loss of active Lck **leads to** failure of TCR/CD3 (and BCR) proximal phosphorylation and ZAP70 activation — i.e., collapse of antigen-receptor signal transduction  
(P 10700239).
5. Failed TCR signaling in the thymus **results in** an arrest of T-cell development at the double-positive → single-positive transition (and reduced DN→DP), impairing both positive and negative selection (demonstrated in mouse;  
P 8666928).
6. Thymic developmental block **leads to** profound peripheral T-cell lymphopenia with non-functional, mitogen-unresponsive T cells (T−)  
(P 10700239).
7. Absent T-cell help, together with intrinsic BCR-signaling impairment, **causes** progressive hypogammaglobulinemia despite normal/increased B-cell numbers (B+)  
(P 10700239).
8. The resulting combined immune failure **leads to** early-infancy recurrent, severe, opportunistic infections and, in one case, lymphoproliferation/B-cell lymphoma — clinical SCID (inferred branch for lymphoma).

**Molecular pathways:** TCR and BCR signaling (Reactome/KEGG "T cell receptor signaling pathway"); the Csk-CD45-Lck regulatory node. **Cellular processes:** thymocyte selection (positive/negative selection), lymphocyte development and activation, TCR-induced apoptosis (defective). **Protein dysfunction:** loss of function of a tandem-domain receptor tyrosine phosphatase; extracellular fibronectin-III lesion abolishes surface expression. **Immune system involvement:** immunodeficiency (not autoimmunity) is the core. **Cell types (CL):** double-positive thymocyte (CL:0000807), T cell (CL:0000084), B cell (CL:0000236), NK cell (CL:0000623). **GO biological processes:** GO:0050852 (TCR signaling), GO:0030217 (T cell differentiation), GO:0006470 (protein dephosphorylation). Upstream = the *PTPRC* lesion and CD45 loss; downstream = defective thymopoiesis, infections, and antibody failure.

## 7. Anatomical Structures Affected

See Finding 9. **Primary organs:** thymus (developmental arrest), bone marrow, spleen, lymph nodes, peripheral blood — the hematolymphoid system. **Secondary/complication organs:** lung (pneumonia) and skin (dermatitis). **Cell/tissue level:** all nucleated hematopoietic cells express CD45; the functional lesion falls hardest on thymocytes and T cells. **Subcellular:** plasma membrane and membrane rafts. Involvement is systemic/bilateral (not lateralized).

## 8. Temporal Development

**Onset:** neonatal-to-early-infancy (recurrent infections from ~2 months). **Onset pattern:** subacute/insidious, becoming acute with severe infection. **Progression:** rapidly progressive and, without immune reconstitution, fatal in early childhood. **Course:** chronic and progressive with acute infectious exacerbations. **Remission:** only treatment-induced, via successful HSCT (durable immune reconstitution). **Critical period:** the neonatal window before infectious/organ complications — pre-symptomatic detection by newborn screening and early transplant define outcome (Finding 6).

## 9. Inheritance and Population

**Inheritance:** autosomal recessive; complete penetrance for biallelic LOF; carriers unaffected. **Expressivity:** somewhat variable across the few cases (e.g., lymphoma in one). **Epidemiology:** ultra-rare — only ~3 unrelated reported families; no disease-specific prevalence. Sits within overall SCID incidence of ~1:46,753–1:58,000 live births by TREC screening. **Consanguinity** is a contributing context (homozygous 6-bp deletion). **No founder effect** established. **Sex ratio** ~1:1 (autosomal). **Carrier frequency** not defined given rarity. No population enrichment documented.

## 10. Diagnostics

**Newborn screening:** TREC assay on dried blood spots (low/absent TRECs in T-lymphopenic infants). **Confirmatory clinical tests:** lymphocyte immunophenotyping by flow cytometry (T–/B+/NK-variable) — **absent surface CD45 on all leukocytes is the disease-specific pointer**; mitogen proliferation assays (absent); serum immunoglobulins (progressive hypogammaglobulinemia). **Genetic testing:** *PTPRC* single-gene sequencing, SCID/IEI gene panels, or WES/WGS; deletion-spanning methods (MLPA/CMA) needed to detect the large-deletion allele. **Differential diagnosis:** other T–B+ SCID (IL2RG/X-linked, JAK3, IL7R

deficiency) — distinguished by preserved surface CD45 in those disorders and by the specific gene defect; also RAG1/2, ADA, and reticular dysgenesis for broader SCID. **Screening:** newborn TREC screening; cascade carrier testing in families.

## 11. Outcome / Prognosis

**Untreated:** uniformly fatal in early childhood. **With HSCT:** curative, with survival strongly dependent on timing of diagnosis. Reported 2-year OS: 71.4% (early/NBS-diagnosed) vs 29.2% (late clinically-diagnosed) SCID; single-center OS 68% overall (93% for NBS-identified) (Finding 6). **Complications:** severe opportunistic infections, failure to thrive, and (in one CD45 case) B-cell lymphoma; transplant-related GVHD and graft failure. **Prognostic factors:** age/timing of diagnosis, pre-transplant infection status, donor matching, stem-cell source, and use of conditioning.

## 12. Treatment

**Definitive:** allogeneic HSCT — the only curative therapy (NCIT: Hematopoietic Cell Transplantation, C15431; Allogeneic Bone Marrow Transplantation). **Supportive/bridging:** immunoglobulin replacement (IVIG; NCIT C509), antimicrobial/antifungal/anti-*Pneumocystis* prophylaxis (e.g., trimethoprim-sulfamethoxazole), protective isolation, irradiated/CMV-safe/leukoreduced blood products, avoidance of live vaccines. **Pharmacogenomics:** conditioning agents (e.g., busulfan) are dose-adjusted; not *PTPRC*-specific. **Advanced/experimental:** gene therapy is investigational for other SCID subtypes but **not established for *PTPRC***; antibody-based (anti-CD45-saporin) non-genotoxic conditioning is under study in models (

**P** 32387109

but not a *PTPRC* therapy. Treatment strategy is genotype-agnostic beyond confirming diagnosis: proceed rapidly to HSCT with optimal supportive care.

## 13. Prevention

**Primary prevention:** not possible (monogenic); genetic counseling and reproductive options (PGT/prenatal testing) reduce recurrence for known-carrier families (25% recurrence risk). **Secondary prevention:** universal newborn TREC screening for presymptomatic detection — the single most impactful intervention (Findings 5–6). **Tertiary prevention:** anti-infective prophylaxis, IVIG, isolation, safe blood products, and avoidance of live vaccines to prevent

complications before HSCT. **Immunization:** live vaccines contraindicated; killed vaccines generally ineffective pre-reconstitution. **Counseling:** genetic counseling and cascade carrier testing.

## 14. Other Species / Natural Disease

**Taxonomy / orthologs:** human *PTPRC* (NCBI Gene 5788); mouse *Ptprc* (NCBI Gene 19264; NCBI Taxon 10090). CD45 is highly conserved across mammals. **Natural disease:** no well-characterized spontaneous CD45-deficiency disease is documented in companion animals in OMIA within this investigation; CD45 biology is conserved and disease models are engineered rather than natural. **Comparative biology:** the mouse knockout closely mirrors the human immunophenotype, supporting strong evolutionary conservation of the CD45→Lck→TCR mechanism (Finding 3). **Zoonotic potential:** none (genetic disorder).

## 15. Model Organisms

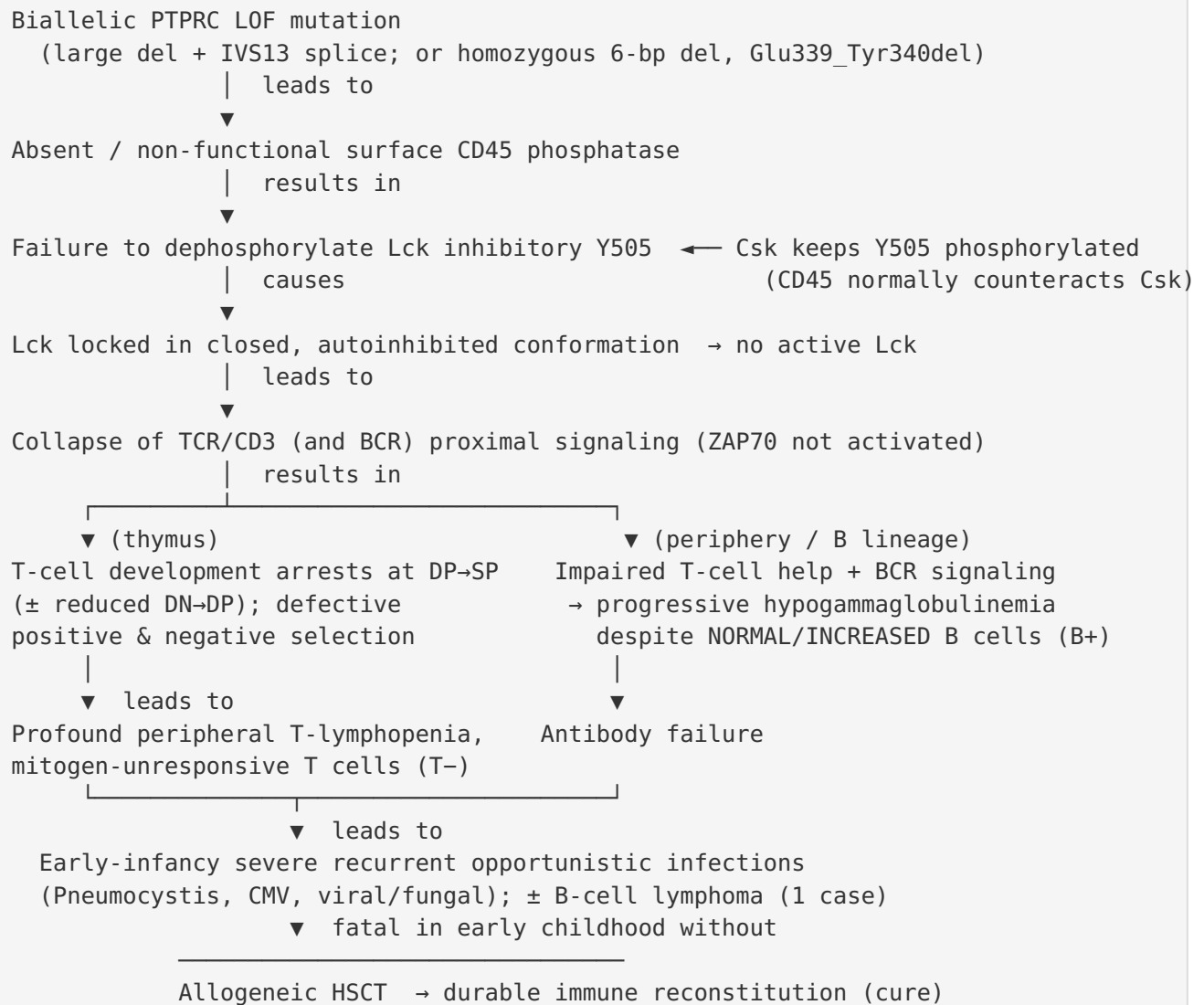
**Mouse (*Mus musculus*)** is the principal model: the CD45/*Ptprc*-null knockout (Byth 1996) recapitulates T–B<sup>+</sup> immunophenotype with a two-stage thymic block, defective negative selection/superantigen deletion, and CD45's role in enhancing positive selection (Finding 3). **Model types:** germline knockout; transgenic CD45RO-reconstitution lines used to dissect isoform effects. **Phenotype recapitulation:** high for the immunophenotype and developmental block. **Limitations:** mouse strain background and the engineered nature of alleles differ from the human private mutations; species differences in isoform usage; the lymphoma predisposition seen in one human patient is not a defining mouse phenotype. **Resources:** MGI (*Ptprc*), IMPC/IMSR for allele availability. In vitro reconstitution systems (supported lipid bilayers;

P 25128530

and cell lines elucidate the CD45/Lck/TCR clustering mechanism.



## Mechanistic Model / Interpretation



CD45 sits at a signaling rheostat: it **activates** Lck by removing the inhibitory phosphate (the dominant net effect in thymocytes) but can also **dephosphorylate the TCR** — a negative role blunted by CD45's spatial exclusion from TCR clusters. In complete deficiency the activating role is lost with no compensation, so the net outcome is a hard block in T-cell development. The mouse knockout, three human patients, and biochemical/reconstitution studies converge on the same causal chain, giving unusually high mechanistic confidence for such an ultra-rare disease.

## Evidence Base

| PMID                     | Title (abbrev.)                                                                      | Role in this report                                                                                |
|--------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <a href="#">10700239</a> | <i>Mutations in the tyrosine phosphatase CD45 gene in a child with SCID</i>          | Founding human case; CD45 regulates Src kinases; T–B+ phenotype, progressive hypogammaglobulinemia |
| <a href="#">11145714</a> | <i>A deletion in the gene encoding the CD45 antigen in a patient with SCID</i>       | Second human case; homozygous 6-bp del; absent surface CD45; locus 1q31-32                         |
| <a href="#">8666928</a>  | <i>CD45-null transgenic mice...</i>                                                  | Mouse knockout recapitulates T–B+ phenotype and two-stage thymic block                             |
| <a href="#">32794043</a> | <i>The role of competing mechanisms on Lck regulation</i>                            | CD45/Csk balance sets Lck activation state                                                         |
| <a href="#">25128530</a> | <i>Phosphatase CD45 both positively and negatively regulates TCR phosphorylation</i> | Dual (activating/inhibitory) CD45 role; spatial exclusion                                          |
| <a href="#">21507955</a> | <i>Disease-associated polymorphism alters CD45 splicing (hnRNPs)</i>                 | Mechanism of C77G — distinct from LOF SCID                                                         |
| <a href="#">17903220</a> | <i>Altered CD45 isoform expression in C77G carriers...</i>                           | C77G = autoimmune/infectious susceptibility, not SCID                                              |
| <a href="#">18312479</a> | <i>PTPRC (CD45) variation and disease association...</i>                             | C77G/A138G isoform-altering susceptibility alleles                                                 |
| <a href="#">42079620</a> | <i>SCID newborn screening, Catalonia</i>                                             | TREC screening; incidence 1:46,753                                                                 |
| <a href="#">40374985</a> | <i>Newborn screening + early treatment in SCID</i>                                   | HCT only cure; 71.4% vs 29.2% 2-yr OS by timing                                                    |
| <a href="#">39900265</a> | <i>HSCT outcome &amp; NBS impact (single center)</i>                                 | NBS-identified OS 93% vs clinical                                                                  |
| <a href="#">32387109</a> | <i>Anti-CD45-saporin conditioning (RAG mice)</i>                                     | Investigational non-genotoxic conditioning (model, not PTPRC therapy)                              |

Supporting mechanistic papers on Lck regulation and TCR clustering (PMIDs 28735895, 33225946, 25127865, 30745330, 25658352, 9052865, 9203971) reinforce the CD45→Lck→TCR axis and thymic selection biology.

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## Limitations and Knowledge Gaps

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- **Extreme rarity:** the human disease rests on only ~3 unrelated patients, so phenotype frequencies, expressivity range, and outcome statistics for *PTPRC*-specific SCID are imprecise; most prognosis/screening data are extrapolated from the broader SCID category.
- **No disease-specific epidemiology:** prevalence/incidence, carrier frequency, and any founder effects are unknown.
- **Genotype–phenotype correlation** cannot be established with so few alleles; the single lymphoma case may be incidental or mechanistically linked (uncertain).
- **Therapy:** no *PTPRC*-specific gene/cell therapy exists; HSCT outcome data are not CD45-specific.
- **Ontology mapping** for a few HPO/GO terms should be verified against current ontology releases before database ingestion.
- **Natural animal disease:** no confirmed spontaneous CD45-deficiency disease documented in OMIA within this investigation.

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## Proposed Follow-up Experiments / Actions

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1. **Query OMIM/ClinVar/HGMD directly** for any additional *PTPRC* SCID cases and variants beyond the three founding families to update the variant table and gnomAD allele-frequency checks.
2. **Curate a differential-diagnosis matrix** for T–B+ SCID (IL2RG, JAK3, IL7R vs *PTPRC*), emphasizing the flow-cytometry discriminator of absent surface CD45.
3. **Assemble ontology cross-checks** (HPO frequency annotations, GO evidence codes, UBERON/CL IDs) against current ontology versions for database ingestion.
4. **Search GTR/GeneReviews** for CD45-specific transplant management protocols and any conditioning-regimen recommendations.

5. **Monitor ClinicalTrials.gov** for gene-editing/HSCT trials that could extend to *PTPRC* SCID, and track anti-CD45-antibody conditioning translation.
  6. **Retrieve MGI/IMPC records** for all *Ptprc* alleles to formalize the model-organism annotation (phenotype recapitulation and limitations).
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*Report compiled from 10 confirmed findings and 26 reviewed papers across a 5-iteration autonomous investigation. Evidence sources: human clinical case reports, mouse knockout studies, in vitro/biochemical reconstitution, and population screening cohorts.*

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